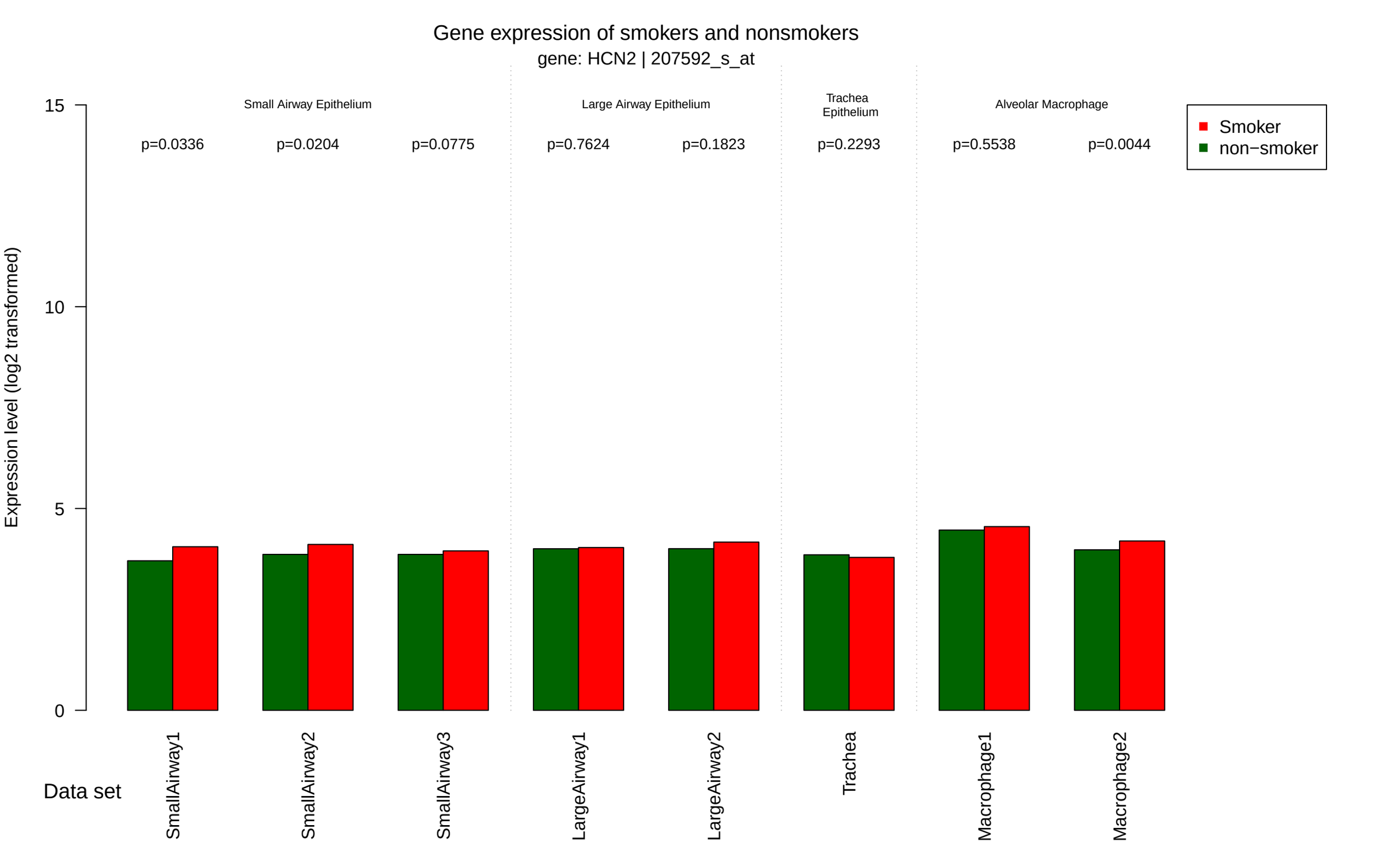
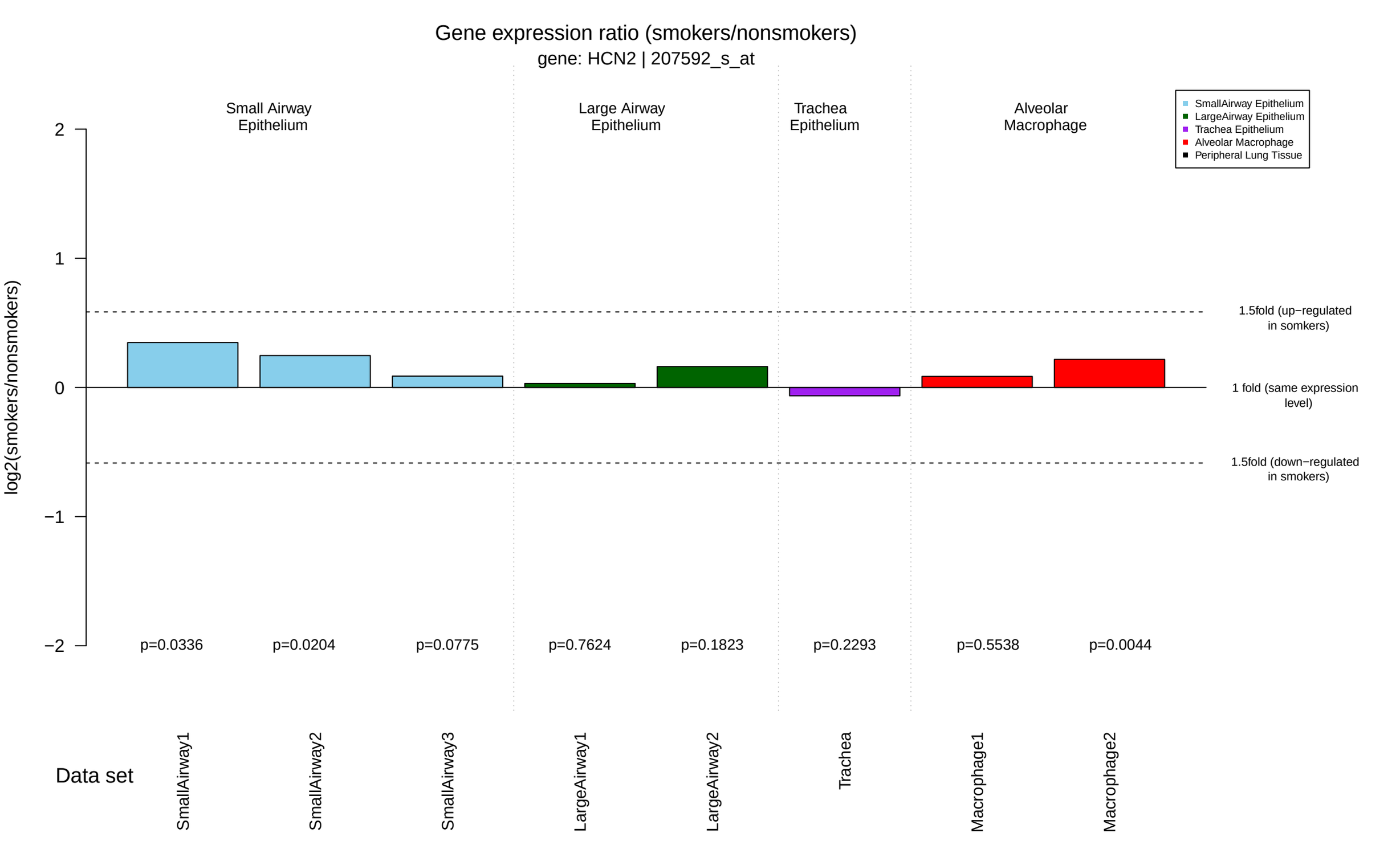
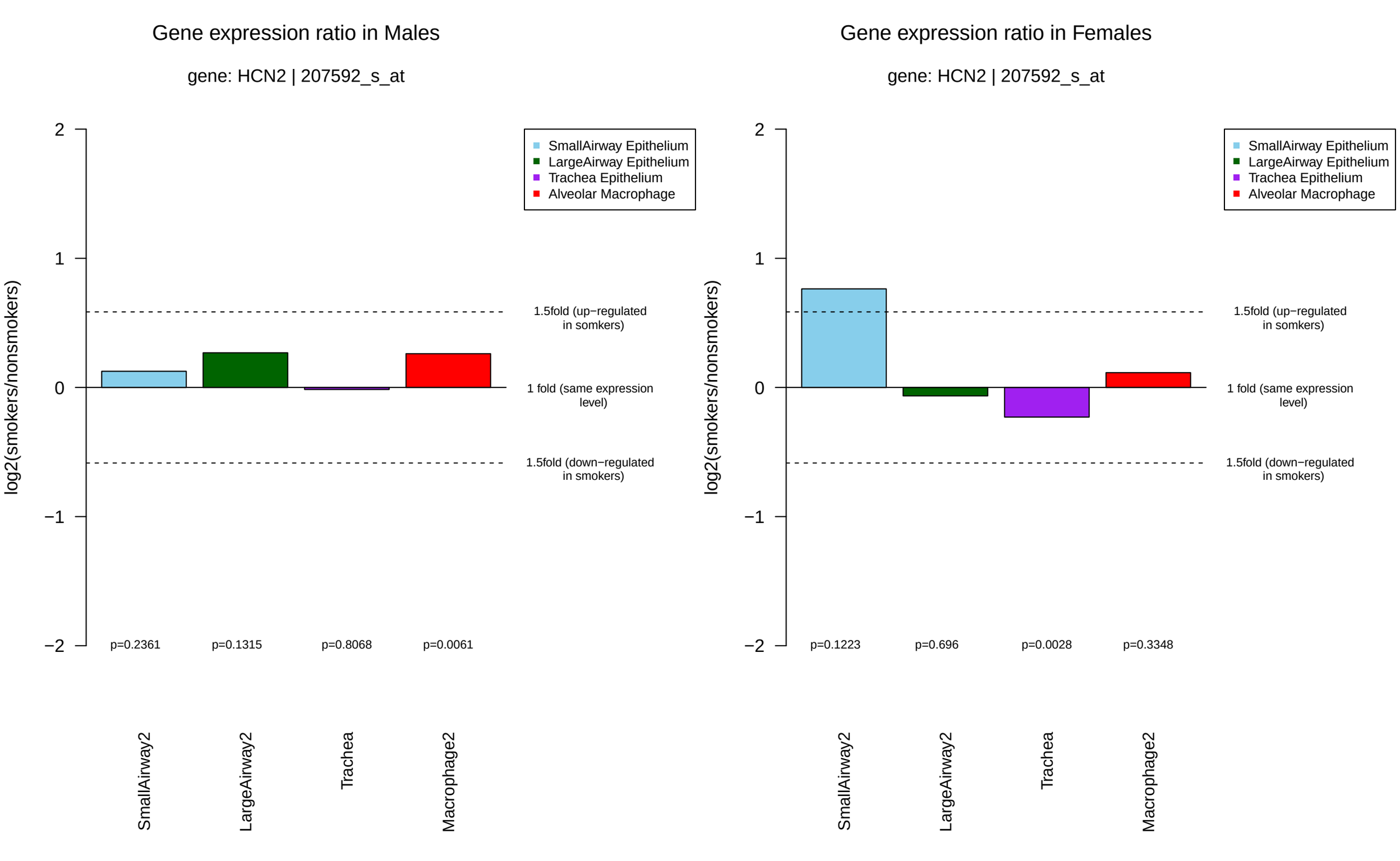
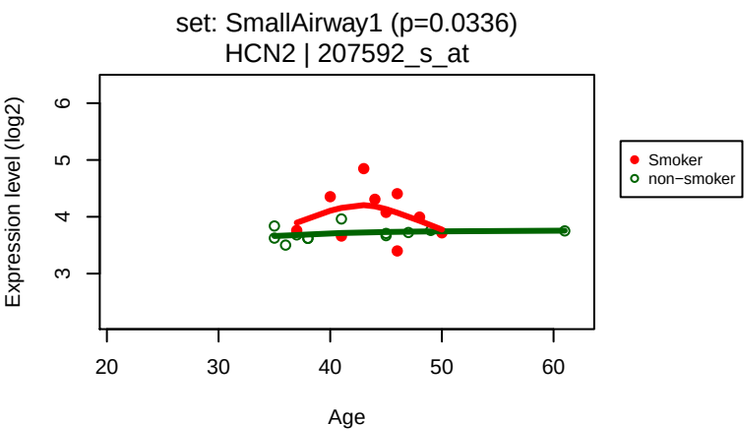
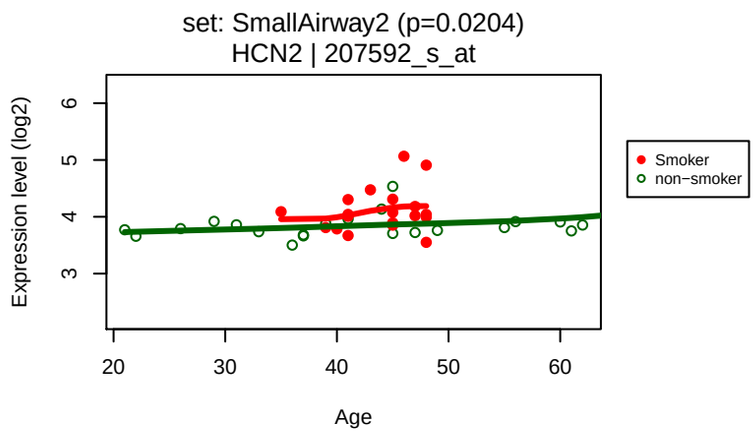
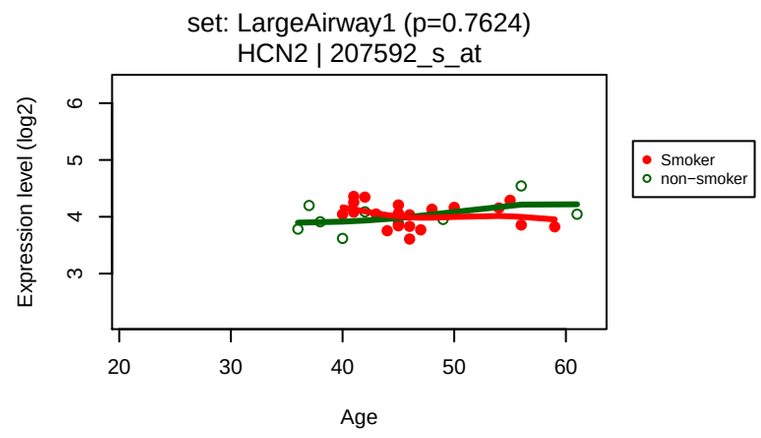

Gene expression of smokers and nonsmokers
gene: HCN2 | 207592_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0336
p=0.0204
p=0.0775
p=0.7624
p=0.1823
p=0.2293
p=0.5538
p=0.0044
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

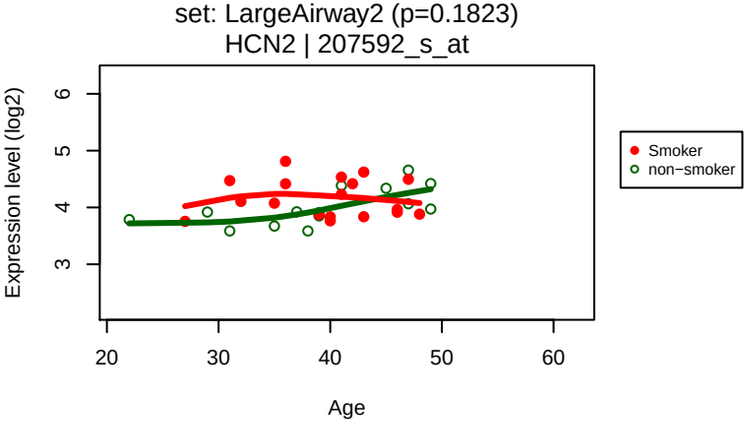

Gene expression ratio (smokers/nonsmokers)
gene: HCN2 | 207592_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0336
p=0.0204
p=0.0775
p=0.7624
p=0.1823
p=0.2293
p=0.5538
p=0.0044
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

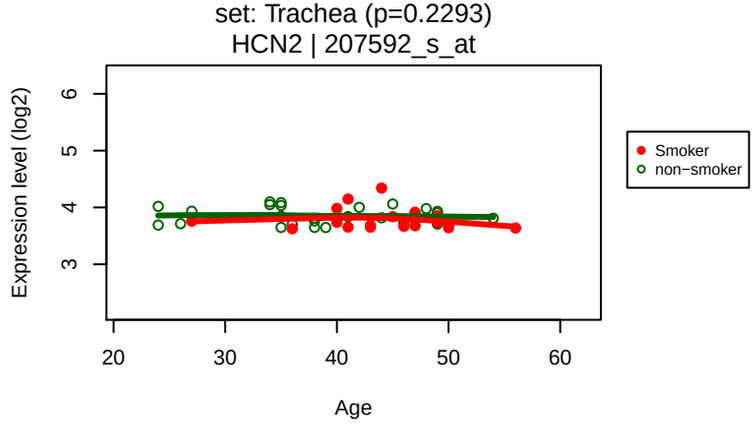

Gene expression ratio in Males
gene: HCN2 | 207592_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2361
p=0.1315
p=0.8068
p=0.0061
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HCN2 | 207592_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1223
p=0.696
p=0.0028
p=0.3348
SmallAirway2
LargeAirway2
Trachea
Macrophage2

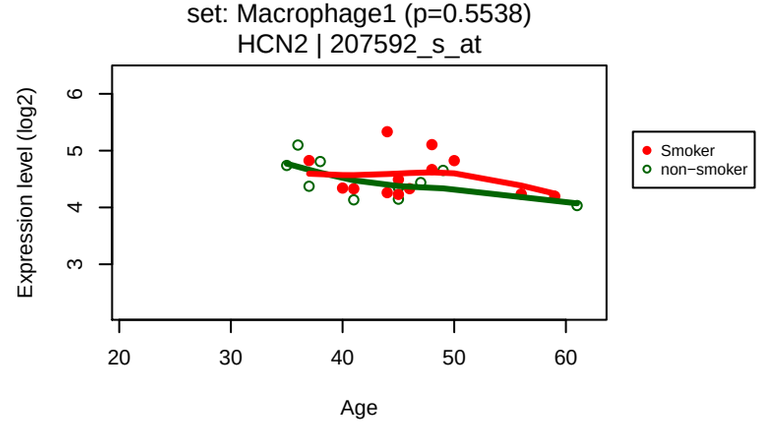

set: SmallAirway1 (p=0.0336)
HCN2 | 207592_s_at
Expression level (log2)
Age
Smoker
non−smoker

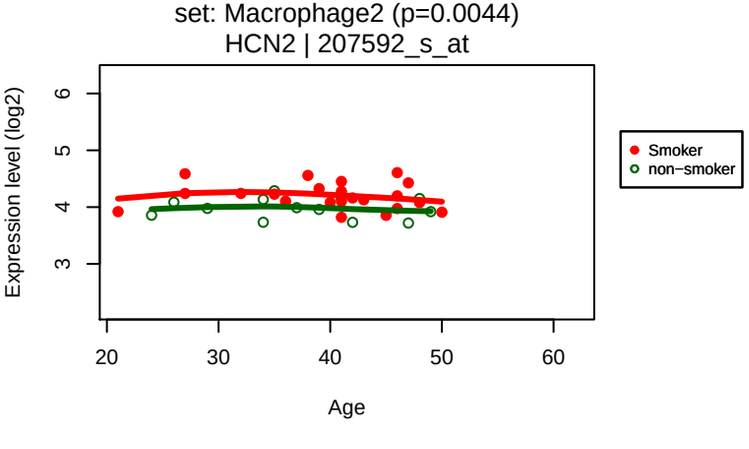

set: SmallAirway2 (p=0.0204)
HCN2 | 207592_s_at
Expression level (log2)
Age
Smoker
non−smoker

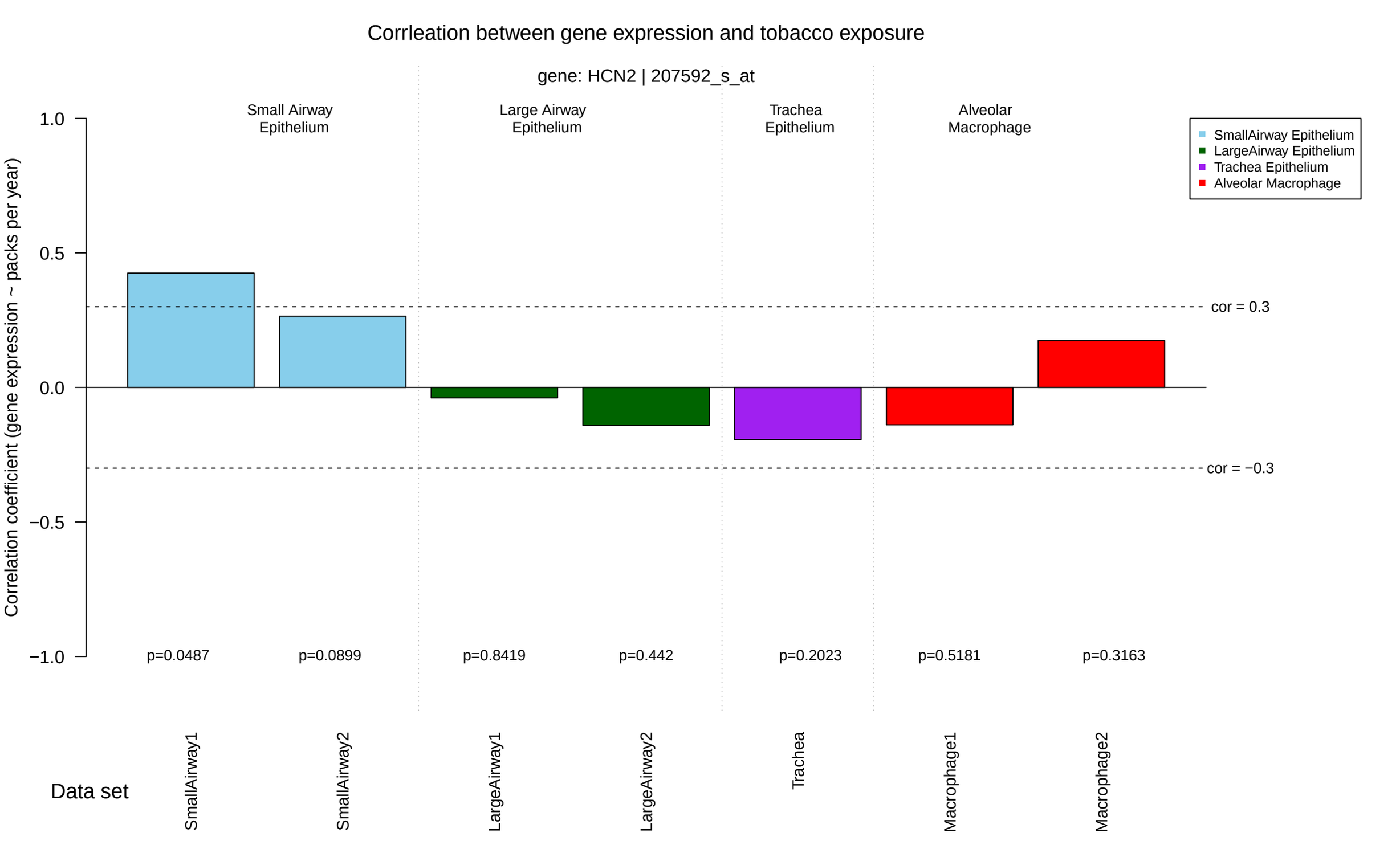

set: LargeAirway1 (p=0.7624)
HCN2 | 207592_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1823)
HCN2 | 207592_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2293)
HCN2 | 207592_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5538)
HCN2 | 207592_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0044)
HCN2 | 207592_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HCN2 | 207592_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0487
p=0.0899
p=0.8419
p=0.442
p=0.2023
p=0.5181
p=0.3163
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2